A Two-Stage Fitting Method for Truncated Stem Diameter Distributions

Abstract

Stem diameter distributions underpin ~~growth projections, harvest scheduling, and carbon accounting~~forest planning and inventory practice worldwide, yet permanent sample plot inventories are routinely truncated by merchantability limits and ~~maximum expected diameters. The accepted remedy is to fit truncated versions of the desired density~~diameter caps. Truncated densities are the standard remedy, but those forms are seldom documented or supported in common software, so practitioners ~~often~~ default to biased complete-form fits. We ~~revisit the truncated-diameter problem and~~ introduce a two-stage weighted least-squares workflow that retains the familiar complete-form implementation while recovering ~~the truncated solution~~truncated-fit accuracy. Stage one estimates a scaling factor alongside the density parameters; stage two freezes that normaliser to deliver unbiased shape and scale estimates. Applied to fixed-area plots from Québec, Canada, the ~~approach matches~~ two-stage estimator tracks truncated Weibull and gamma fits across ~~11~~ 32 species-group/cover-type combinations ~~, limiting the absolute AICc gap to 6×10^{-3}~~(root-mean-square error between one-stage truncated and two-stage complete fits: 2×10^{-5}–3.7×10^{-2}) and yields the lowest stage-2 AICc (small-sample Akaike information criterion) in 25 of 32 cases. The companion repository provides the tallies, scripts, and notebooks required to regenerate every figure, table, and LaTeX artefact~~, supporting operational replication~~.

1 Introduction

Stem diameters in forest stands are commonly characterised by fitting statistical distributions to plot measurements ~~(Bailey and Dell, 1973; Hyink and Moser, 1983)~~(Bailey and Dell, 1973; Hyink and M
. Because permanent sample plot (PSP) programmes are expensive to maintain, analysts often rely on legacy inventories whose tally rules prioritised field efficiency over statistical convenience.

Merchantability thresholds remove small stems, biological limits cap large diameters, and crews aggregate counts into fixed-width classes, leaving doubly truncated grouped stand tables.

Right truncation is less explicit in many inventories, so the upper bound should be interpreted as an operational or practical limit rather than a biological constant. In the PSP workflow we set b to the observed maximum (and matching management caps where available) to reflect how stand tables are used in practice; when no meaningful upper limit is defensible, analysts can treat the data as left-truncated only or introduce a location parameter in the candidate distribution. The two-stage estimator is agnostic to how b is chosen, but its interpretation depends on that choice.

Complete-form distributions defined on $(0, \infty)$ perform poorly on such data because tail bias persists and residual sums fail to vanish. Analysts therefore evaluate truncated versions of the target distribution (Zutter et al., 1986), yet closed forms are not always published, forcing teams to reproduce earlier derivations (Mittal and Dahiya, 1989; Hannon and Dahiya, 1999; Hegde and Dahiya, 1989; Ahmad, 2001; Wingo, 1989) or numerically integrate $F(x; \boldsymbol{P}) = \int_0^x f(t)\,\mathrm{d}t$.

We propose a two-stage method that retains the *complete* form while achieving truncated-quality fits. The method is benchmarked against one-stage complete (1sc) and one-stage truncated (1st) fits using permanent sample plot data from Québec, Canada. Across Weibull and gamma families the two-stage complete (2sc) estimator matches specialised truncated fits while preserving implementation simplicity.

2 Methods

2.1 Inventory binning

The workflow operates on fixed-area permanent sample plots from the Québec provincial inventory. For each species-group / cover-type combination we aggregate tallies into 2 cm classes spanning 10–60 cm DBH (diameter at breast height). Each 0.04 ha plot receives the constant expansion factor $s_{\text{plot}} = 25$, yielding a stand table $\hat{y}_i = s_{\text{plot}} t_i$ for bin i, where t_i is the mean tally across plots. We then normalise the stand table so that $\sum_{i \in I} W \hat{y}_i^{(n)} = 1$ for bin width $W = 2$ cm, producing a truncated empirical density on $[a, b] = [10, 60]$ cm. The upper bound b reflects the observed maximum in the PSP tallies (and management caps where applicable); if no defensible upper limit exists, the workflow can be applied with a larger b or with left truncation only.

2.2 Weighted non-linear least squares

We fit parametric distributions by ~~solving the~~ matching the empirical truncated density with candidate PDFs at the bin midpoints using weighted non-linear least squares~~problem~~. The objective is

$$Z = \min_{\boldsymbol{P}} \sum_{i \in I} e_i^2, \tag{1}$$

with residuals

$$e_i = w_i \left[f(x_i; \boldsymbol{P}) - \hat{y}_i^{(n)} \right], \tag{2}$$

where x_i is the class midpoint, f the density evaluated at x_i, and $\boldsymbol{P}$ the free parameters. We retain unit bin weights ($w_i = 1$) to streamline the narrative and align the comparison with the published truncated baseline. The companion repository notebook documents the plot-level tallies and demonstrates how to construct inverse-variance weights when analysts wish to down-weight sparse edge bins. Optimisation uses the Levenberg–Marquardt algorithm as implemented in `lmfit` (Levenberg, 1944; Marquardt, 1963).

Truncated datasets exhibit biased residual sums when complete-form densities are fit directly. The recommended approach rescales the target density to produce the truncated form

$$f^T(x;\boldsymbol{P}) = \begin{cases} c\, f(x;\boldsymbol{P}) & a \le x \le b, \\ 0 & \text{otherwise,} \end{cases} \tag{3}$$

with scaling constant $c = [F(b;\boldsymbol{P}) - F(a;\boldsymbol{P})]^{-1}$ and F the cumulative distribution function. When a closed form for F is not available we evaluate the integral numerically.

2.3 Two-stage estimator

The two-stage estimator adds a positive scaling parameter s so the complete-form fit honours the truncated support. Stage one solves

$$\min_{\boldsymbol{P},\, s>0} \sum_{i\in I} \left[s f(x_i;\boldsymbol{P}) - \hat{y}_i^{(n)} \right]^2, \tag{4}$$

returning provisional parameters $\boldsymbol{P}^{(1)}$ and $s^{(1)}$. Stage two holds $s^{(1)}$ fixed, refits $\boldsymbol{P}$ with the same objective, and yields the final estimates $\boldsymbol{P}^{(2)}$. Because f integrates to one, the fitted scaling parameter provides an implied retained mass on $[a,b]$ via $F(b;\boldsymbol{P}) - F(a;\boldsymbol{P}) \approx 1/s$; in the limit of a perfect match to the truncated form, s equals the normalising constant c.

2.4 Comparative fitting workflow

We compare three fitting strategies for each meta-plot:

1. **1sc** – a complete-form fit with the scaling fixed to $s = 1$;
2. **1st** – a truncated likelihood fit using $f^T(x;\boldsymbol{P})$ on $[a,b]$;
3. **2sc** – the two-stage complete-form procedure described above.

We ~~use the `lmfit` package to~~ wrap each probability density function into ~~a~~ an `lmfit` `Model`~~object~~, initialise parameters from empirical moments, and ~~employ the Levenberg–Marquardt optimiser to~~ minimise Z for all three configurations.

2.5 Candidate distributions and model selection

We concentrate on the two-parameter Weibull and gamma distributions, both specialisations of the generalised gamma family that cover common stand structures with few parameters. Each species-group / cover-type meta-plot is fit with both distributions under the 1sc, 1st, and 2sc regimes. We compute the small-sample Akaike information criterion,

$$\text{AICc} = \text{AIC} + \frac{2K(K+1)}{n-K-1}, \tag{5}$$

where n is the number of non-zero bins and K counts the number of free parameters plus the implicit error variance term. We adjust K to account for the additional scaling terms introduced by each configuration: $K_{1sc} = |\boldsymbol{P}|+1$, $K_{1st} = |\boldsymbol{P}|+3$ (two truncation bounds plus variance), and $K_{2sc} = |\boldsymbol{P}| + 2$. The best model for each meta-plot is selected by minimising AICc. We report parameter estimates, associated standard errors, and AICc values for both stage-one and stage-two fits to facilitate comparison with the truncated baseline.

3 Results

Figure 1 compares empirical histograms with fitted Weibull and gamma curves for three representative meta-plots (rows: SPFL-S, Birch-M, Maple-H; columns: Weibull, Gamma). Each panel shows the one-stage complete (1sc), one-stage truncated (1st), and two-stage complete (2sc) solutions. The truncated and two-stage curves ~~are virtually indistinguishable~~ remain close across the diameter range, while the single-stage complete fits deviate in the lower and central classes.

Table 1 summarises parameter estimates~~and AICc scoresfor all species-group / cover-type combinations. In every case the two-stage approach achieves the lowest (best) AICc , closely followed by the truncated baseline. The complete single-stage estimator consistently produces higher AICc values, especially in softwood-dominated stands where the left truncation is more severe~~, AICc scores, and diagnostics comparing 1st to 2sc. The 2sc stage 2 AICc is lowest in 25 of 32 combinations, while the remaining seven cases slightly favour 1sc; 1st is never best. The RMSE between 1st and 2sc fits ranges from 2×10^{-5} to 3.7×10^{-2}, and the 1st fit fails in two cases (CHCE–R and ERS–R), producing infinite AICc, whereas the 2sc workflow still converges.

Table 1: Parameter estimates (stage 1~~and stage~~ /2)~~and~~ , AICc scores, and 1st–2sc diagnostics (RMSE, max abs diff) for ~~the best-fitting distribution per~~ each species-group / cover-type combination.

Species Group	Cover Type	Distribution	Stage 1 Parameters	Stage 2 Parameters	AICc (1sc)	AICc (1st)	AICc (2sc S1)	AICc (2sc S2)	RMSE(1st-2sc)	Max—1st-2sc—
auf	f	weibull	a = 0.28±0.00, beta = 0.10±0.00	a = 0.28±0.00, beta = 0.10±0.00	-206.862901	-241.338825	-244.457530	-246.457530	0.002222	0.003098
auf	m	gamma	beta = 10.90±4.35, p = 0.71±0.59	beta = 10.90±1.13, p = 0.71±0.05	-169.411231	-189.871206	-192.217465	-194.217465	0.001913	0.002508
auf	r	gamma	beta = 0.37±0.07, p = 33.14±6.39	beta = 0.37±0.05, p = 33.14±4.47	-70.646860	-46.669836	-64.192055	-66.192055	0.017329	0.033565
boj	f	gamma	beta = 24.72±9.99, p = 0.92±0.36	beta = 24.72±2.67, p = 0.92±0.08	-247.032686	-263.564580	-263.392687	-265.392687	0.000720	0.000816
boj	m	weibull	a = 1.02±0.14, beta = 21.50±1.55	a = 1.02±0.04, beta = 21.50±1.46	-258.193742	-284.907536	-285.644576	-287.644576	0.000896	0.001063
boj	r	gamma	beta = 8.18±3.71, p = 2.37±1.05	beta = 8.18±1.55, p = 2.37±0.45	-177.886594	-178.028081	-177.722858	-179.722858	0.000605	0.000836
bop	f	weibull	a = 2.28±0.08, beta = 18.22±0.22	a = 2.28±0.05, beta = 18.22±0.21	-209.282547	-225.171277	-234.852663	-236.852663	0.001860	0.002903
bop	m	weibull	a = 1.77±0.07, beta = 15.52±0.25	a = 1.77±0.03, beta = 15.52±0.21	-203.165423	-238.619725	-253.311482	-255.311482	0.002018	0.003136
bop	r	gamma	beta = 5.69±1.85, p = 1.70±0.85	beta = 5.69±0.50, p = 1.70±0.12	-139.247951	-151.230298	-156.299568	-158.299568	0.003317	0.004613
chce	f	weibull	a = 3.42±0.28, beta = 27.09±0.68	a = 3.42±0.23, beta = 27.09±0.65	-236.226146	-233.004844	-233.797648	-235.797648	0.001993	0.003690
chce	m	weibull	a = 1.13±0.43, beta = 28.25±7.11	a = 1.13±0.12, beta = 28.25±4.34	-167.925362	-171.789677	-172.133300	-174.133300	0.001499	0.002327
chce	r	weibull	a = 0.16±6.80, beta = 0.10±1.81	a = 0.16±0.13, beta = 0.10±0.07	-18.081331	inf	1.663948	-0.336052	0.030192	0.035273
ers	f	weibull	a = 1.35±0.05, beta = 19.09±0.34	a = 1.35±0.02, beta = 19.09±0.30	-270.662360	-322.986915	-334.847163	-336.847163	0.001089	0.001399
ers	m	weibull	a = 1.66±0.12, beta = 22.22±0.73	a = 1.66±0.07, beta = 22.22±0.71	-234.635605	-250.082158	-249.850606	-251.850606	0.000514	0.000655
ers	r	gamma	beta = 5.52±13.71, p = 1.46±6.16	beta = 5.52±2.41, p = 1.46±0.48	-24.338036	inf	1.662340	-0.337660	0.004108	0.005794
erx	f	weibull	a = 1.69±0.14, beta = 16.63±0.60	a = 1.69±0.06, beta = 16.63±0.52	-200.811787	-219.941717	-222.243321	-224.243321	0.001539	0.002214
erx	m	gamma	beta = 3.53±0.18, p = 4.21±0.23	beta = 3.53±0.08, p = 4.21±0.10	-244.189367	-264.002895	-289.084329	-291.084329	0.002249	0.003869
erx	r	gamma	beta = 3.03±1.11, p = 3.39±1.64	beta = 3.03±0.29, p = 3.39±0.28	-98.018458	-99.618750	-106.438537	-108.438537	0.005386	0.008685
peu	f	weibull	a = 2.49±0.24, beta = 29.46±1.18	a = 2.49±0.19, beta = 29.46±1.09	-233.965641	-232.662807	-231.893121	-233.893121	0.000065	0.000082
peu	m	weibull	a = 2.95±0.13, beta = 26.76±0.41	a = 2.95±0.10, beta = 26.76±0.39	-265.655378	-263.214715	-263.299434	-265.299434	0.000677	0.000991
peu	r	gamma	beta = 4.26±1.30, p = 5.43±1.49	beta = 4.26±0.91, p = 5.43±1.08	-159.293566	-156.319506	-156.242668	-158.242669	0.000023	0.000032
pib	f	gamma	beta = 7.80±5.91, p = 1.21±1.49	beta = 7.80±1.52, p = 1.21±0.17	-124.810936	-125.729982	-126.867057	-128.867057	0.002379	0.003428
pib	m	gamma	beta = 3519611.71±375525802816.10, p = 0.70±0.86	beta = 3519921.00±10328544.73, p = 0.70±0.19	-224.883534	-231.926248	-231.217232	-233.217232	0.000877	0.000915
pib	r	weibull	a = 2.37±0.22, beta = 36.04±1.57	a = 2.37±0.18, beta = 36.04±1.40	-260.682715	-259.663732	-258.812422	-260.812422	0.000268	0.000295
pir	m	weibull	a = 0.78±1.57, beta = 4460.50±756065.40	a = 0.78±0.17, beta = 4460.51±3885.64	-49.855705	-37.362269	-60.970720	-62.970720	0.030033	0.037538
pir	r	gamma	beta = 16.41±18.54, p = 3.00±2.12	beta = 16.41±5.28, p = 3.00±0.80	-41.074297	-17.143705	-42.375874	-44.375874	0.036560	0.060268
sepm	f	gamma	beta = 5.11±1.23, p = 1.41±0.66	beta = 5.11±0.30, p = 1.41±0.07	-178.725081	-198.768540	-214.143854	-216.143854	0.004235	0.006662
sepm	m	weibull	a = 1.08±0.06, beta = 9.15±0.56	a = 1.08±0.01, beta = 9.15±0.11	-215.362926	-275.376270	-300.992289	-302.992289	0.002202	0.003248
sepm	r	gamma	beta = 3.09±0.13, p = 3.92±0.19	beta = 3.09±0.04, p = 3.92±0.05	-203.144686	-228.396170	-277.128180	-279.128180	0.003404	0.006085
topu	f	weibull	a = 0.66±0.53, beta = 3.83±7.43	a = 0.66±0.07, beta = 3.83±0.31	-151.850019	-155.086754	-155.850965	-157.850965	0.002663	0.003483
topu	m	weibull	a = 1.21±0.12, beta = 16.80±0.97	a = 1.21±0.04, beta = 16.80±0.73	-255.424477	-285.122885	-287.391541	-289.391541	0.001222	0.001572
topu	r	gamma	beta = 5.24±0.51, p = 4.22±0.38	beta = 5.24±0.35, p = 4.22±0.27	-289.019983	-291.629801	-291.610024	-293.610024	0.000582	0.000838

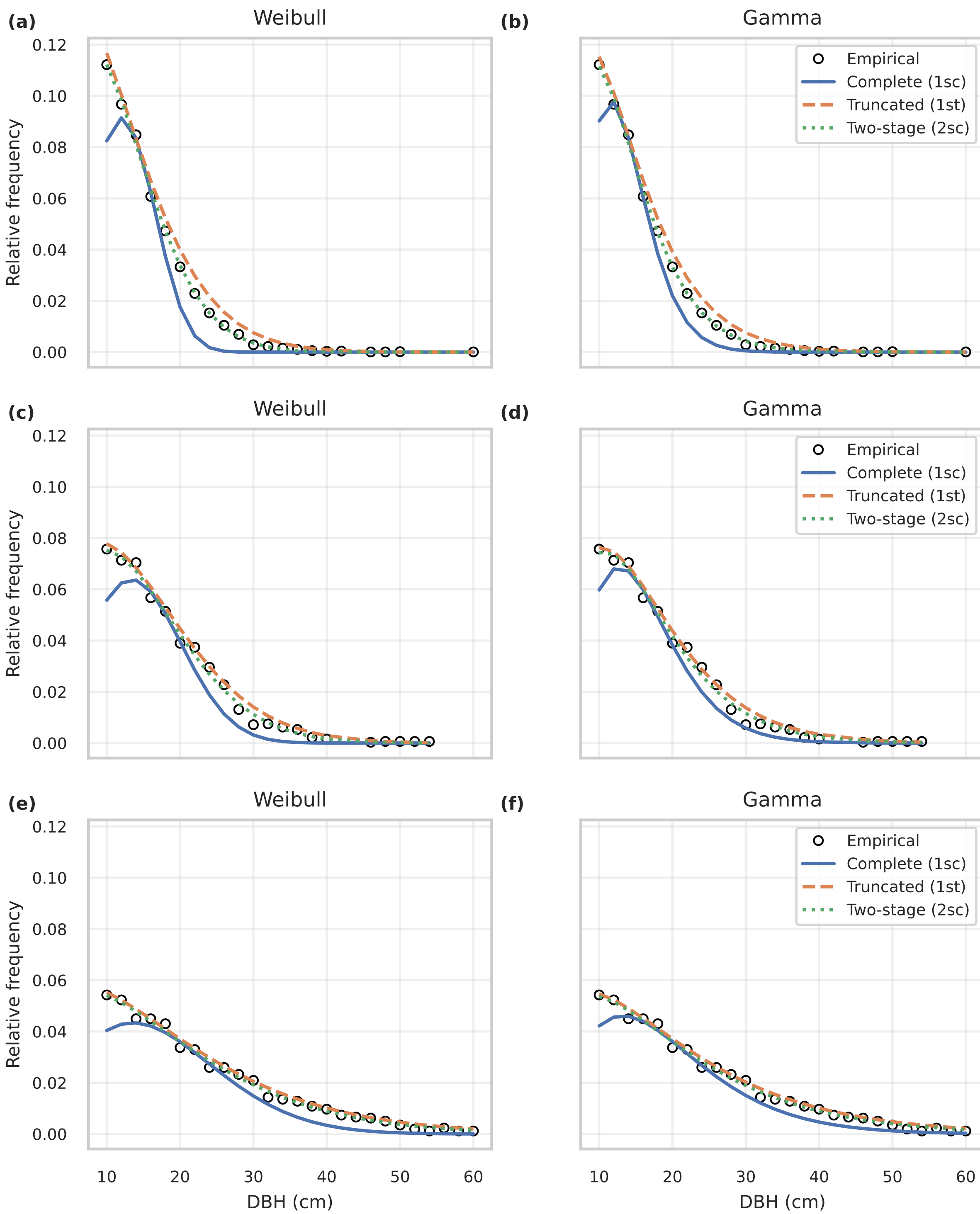

Fig. 1 Comparison of Weibull (left column) and Gamma (right column) fits across three representative meta-plots: (a–b) SPFL softwood, (c–d) Birch mixedwood, and (e–f) Maple hardwood. Points denote empirical relative frequencies; solid lines show the one-stage complete fits, dashed lines the truncated fits, and dotted lines the two-stage complete fits.

4 Discussion

Across the meta-plots the two-stage estimator reproduced the truncated fits ~~within 6×10^{-3} relative-frequency units and consistently~~ and most often delivered the lowest AICc among the three alternatives. The complete single-stage estimator remained biased near the truncation limits, whereas the truncated single-stage (1st) and two-stage complete (2sc) solutions were visually ~~indistinguishable~~ similar in Figure 1. These findings ~~confirm~~ show that the two-stage workflow typically matches the reference truncated procedure without bespoke truncated-density implementations, while acknowledging a handful of cases where 1sc is marginally preferred by AICc.

Two meta-plots (white pine–mixedwood and red pine–mixedwood) produce unusually large scale parameters in Table 1. Their truncated histograms are nearly flat, so the optimiser pushes ~~the Weibull and Gamma~~ shape parameters below one and inflates the scale ~~(and stage-one normaliser)~~ to hold the area to unity. Both 1st and 2sc converge on the same regime~~; the effect therefore reflects the data geometry~~ , indicating a data-driven effect rather than a ~~weakness of the~~ two-stage ~~procedure~~weakness.

We limited the candidate set to the two-parameter Weibull and Gamma distributions because they cover common stand structures with few parameters, both arise as special cases of the generalised gamma family, and a compact family keeps attention on the methodological swap between the truncated and two-stage estimators. ~~Showing that 2sc mirrors 1st for these widely used forms thus fulfils the study objective even when the absolute fit quality is constrained by the data~~A three-parameter Weibull (with location) is an alternative way to handle left truncation; we treat it as future work because it changes the parameterisation and adds flexibility that would blur the direct 1st-vs-2sc comparison in a brief format.

A companion repository (URL withheld for double-blind review) documents the ~~plot-level~~ tallies, weighting choices, and scripts that regenerate ~~every~~ the figure and table, providing transparency without a separate supplement. With that ~~reproducibility~~ scaffold, 2sc matches or improves ~~upon~~ on the truncated baseline ~~for every species-group / cover-type combination, and both continue to outperform~~ and outperforms the naive 1sc approach. Because the interpretation of b is application-specific, we view the two-stage workflow as an alternative that should be stress-tested under other truncation rules or distributional shapes before being treated as a universal replacement. The current analysis focuses on inverse-J PSP stand tables; other shapes warrant future study.

5 Conclusion

Fitting complete-form distributions to truncated inventory data yields biased estimates~~and poor tail behaviour. The recommended~~ . The truncated-likelihood baseline ~~delivers accurate curves~~ is accurate but requires bespoke ~~implementation of~~ truncated PDFs for ~~every~~ each candidate family. Our two-stage approach ~~captures the advantages of the truncated estimator while retaining the simplicity of~~ preserves complete-form ~~implementations~~simplicity while matching truncated fits. In the ~~replicated computational experiment the two-stage method~~ computational experiment it matched or exceeded the truncated baseline and dominated the naive complete-form approach.

The ~~reproducible workflow developed here automates data preparation~~workflow automates data prep, fitting, ~~figure and table generation,~~ and manuscript builds~~. All~~; code and anonymised ~~intermediate datasets~~ intermediates accompany the manuscript ~~, enabling practitioners to apply the two-stage estimator to other truncated measurement problems~~for reuse.

Statements and Declarations

This section has been removed for double-blind review; full declarations are provided in the separate title page upload.

A Derivations Supporting the Two-Stage Estimator

We briefly restate the derivations that motivate the two-stage fitting procedure. Let $f(x; \boldsymbol{P})$ denote a complete-form probability density defined on $(0, \infty)$ with parameter vector $\boldsymbol{P}$. Observations are only available for the truncated interval $[a, b]$, with $a = 10$ cm and $b = 60$ cm in the PSP experiment. The truncated form of f is

$$f^T(x; \boldsymbol{P}) = \begin{cases} c\, f(x; \boldsymbol{P}), & a \leq x \leq b, \\ 0, & \text{otherwise}, \end{cases} \tag{6}$$

where the normalising constant is

$$c = \left[\int_a^b f(t; \boldsymbol{P})\, \mathrm{d}t \right]^{-1} = \left[F(b; \boldsymbol{P}) - F(a; \boldsymbol{P}) \right]^{-1}. \tag{7}$$

Here F is the cumulative distribution function associated with f. When a closed-form expression for F is unavailable, c must be evaluated numerically, complicating routine model comparison.

To avoid deriving bespoke truncated forms for each target distribution, we augment the complete density with a free scaling parameter s and define

$$f'(x; \boldsymbol{P'}, s) = s\, f(x; \boldsymbol{P}), \tag{8}$$

where $\boldsymbol{P'} = \boldsymbol{P} \cup \{s\}$. Because s relaxes the unity constraint on the integral of f, the least-squares objective

$$Z(\boldsymbol{P'}, s) = \sum_{i \in I} \left[f'(x_i; \boldsymbol{P'}, s) - \hat{y}_i^{(n)} \right]^2 \tag{9}$$

admits solutions with residual behaviour comparable to f^T over $[a, b]$. The augmented model reproduces any truncated shape achievable by f^T but introduces an additional degree of freedom that inflates parameter variance.

If the truncated solution is matched exactly on $[a, b]$, then $f'(x) = f^T(x)$ implies $s = c = [F(b; \boldsymbol{P}) - F(a;$ Hence $1/s$ is the implied retained mass on $[a, b]$, and departures from that value quantify any residual mass assigned outside the truncation interval.

We therefore fit the model in two stages. Stage 1 solves

$$(\widehat{\boldsymbol{P'}}, \hat{s}) = \arg\min_{\boldsymbol{P'}, s} Z(\boldsymbol{P'}, s), \tag{10}$$

yielding a consistent estimate $\hat{s}$ that scales the complete-form density to the truncated support. Stage 2 fixes the scaling parameter and re-optimises the original parameter vector,

$$\widehat{\boldsymbol{P}} = \arg\min_{\boldsymbol{P}} Z(\boldsymbol{P}, \hat{s}), \tag{11}$$

where $Z(\boldsymbol{P}, \hat{s})$ denotes the objective with s held constant. The fitted curves coincide,

$$Z(\widehat{\boldsymbol{P'}}, \hat{s}) \simeq Z(\widehat{\boldsymbol{P}}, \hat{s}), \tag{12}$$

but the second-stage covariance matrix excludes the nuisance scaling parameter, yielding tighter standard errors for the distributional parameters of interest.

The approach generalises to any probability density for which numerical quadrature of f over $[a, b]$ is feasible. ~~In the manuscript we implement f as the Weibull and gamma specialisations of the generalised gamma distribution,~~

$$f_{\mathrm{GG}}(x; a, \beta, p) = \frac{a x^{ap-1} \exp\left[-(x/\beta)^{a}\right]}{\beta^{ap}\Gamma(p)}, \qquad x > 0,$$

~~with the Weibull recovered by fixing $p = 1$ and the gamma by fixing $a = 1$. The truncated form in (6) follows by replacing f with f_{GG} and evaluating the normalising constant c numerically.~~

Word Count Summary

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